

Urban Traffic Carbon Emission Prediction: New York City Case Study

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<https://github.com/LinneaLiny/Urban-Traffic-Carbon-Emission-Prediction.git>

Background and Purpose

In the context of the current rapid development of urbanization, transportation carbon emissions have become an issue that cannot be ignored in the sustainable development of cities. As one of the busiest cities in the world, New York City's complex transportation system not only supports the daily travel of millions of citizens but also quietly accumulates huge environmental pressure. The interaction between air pollution, road congestion, and carbon emissions constitutes a dynamic and urgent challenge. For this reason, I hope to explore data-driven solutions through this project, combining my long-term attention to environmental protection issues with training in data science methods, and try to build a bridge between "environmental protection-machine learning prediction-intervention".

The core question of this study is: *How to accurately predict the short-term carbon emission levels in different areas of New York City based on multi-source traffic and environmental data, and provide a decision-making basis for the future dynamic path optimization system?* At this stage, I focus on building a high-resolution, spatiotemporally mapped carbon emission prediction model to capture the spatial distribution and time-varying laws of carbon emissions in the city. On this basis, in the future, we will explore integrating the prediction results into the actual navigation system to promote low-carbon path recommendations and real-time carbon emission intervention.

Data Source

In order to accurately model the carbon emission levels of transportation in different areas of New York City, I selected multi-source heterogeneous data covering three dimensions: travel, environment, and meteorology. These data not only have strong explanatory power but also have public accessibility and high temporal resolution.

The travel data comes from the NYC TLC Trip Record, which covers detailed trip information of yellow taxis and high-capacity online ride-hailing cars (such as Uber and Lyft), including the time and location of boarding and alighting, travel distance and time, area code, and estimated travel speed. These data are updated daily and made public. They are highly timely and fine-grained, and are an important basis for carbon emission estimation.

The environmental data mainly comes from the AirData platform of the US Environmental Protection Agency (EPA). I extracted PM2.5 concentrations and other pollutant data from multiple monitoring points in New York City to characterize the air quality background during the travel period. Due to the large proportion of missing pollutants such as NO₂ and O₃, we chose to retain only PM2.5 as a representative variable.

The meteorological data comes from historical weather observation data, including the average daily temperature in each region. Temperature is included in the modeling system because it has a significant impact on vehicle energy consumption and pollution generation. All external data are matched to travel data by timestamp and region name.

For the sake of analysis accuracy and interpretability, I focus on the five boroughs of New York City: Manhattan, Brooklyn, Queens, Bronx, and Staten Island. These areas are not only the most densely populated areas in the city, but also have complete regional division information, which helps to characterize emission distribution patterns in space. Outlying areas such as EWR (Newark Airport) have sparse travel data and lack consistency, so they are not included in the analysis.

Data structure

In terms of data structure, both datasets(yellow taxi and high-capacity online ride-hailing cars) contain 13 core variables. *pickup_datetime* and *dropoff_datetime* record the time of pick-up and drop-off, which can be used to calculate the trip duration and identify traffic peaks; *trip_distance* (or *trip_miles*) represents the ground driving distance (unit: miles), but there are abnormally long distance values; *PULocationID* and *DOLocationID* are the pickup and drop-off area codes designated by New York City, and *pickup_borough* and *dropoff_borough* are the corresponding administrative district names, which are convenient for spatial analysis.

Trip_time (unit: seconds) and *avg_speed_mph* (unit: miles/hour) reflect the travel time and average speed respectively, and some records have extremely abnormal values. *RatePerDistance* represents the cost per unit distance, which is used to measure the economy of travel. This variable is missing in some records. *CarbonEmission_g* is the carbon emissions (unit: grams) estimated based on the trip characteristics, which is an important indicator for environmental impact assessment. *PM2.5* records the concentration of fine particulate matter (unit: $\mu\text{g}/\text{m}^3$) in the corresponding area during the trip, which is used to characterize air quality, but some values are unreasonable; *avg_temp* represents the average temperature (unit: Fahrenheit) during the trip, which is related to energy consumption and emissions.

Variable types include timestamp (POSIXct), numeric types (such as distance, speed) and categorical types (such as borough), and basic statistics show a certain degree of skewness and missingness.

Preliminary Analysis

Univariate Analysis

To gain a more comprehensive understanding of the characteristics of online ride-hailing trips, I conducted a univariate exploratory analysis of key variables. First, carbon emissions showed an obvious right-skewed distribution, with carbon emissions for most trips concentrated below 2,000 grams, and relatively few records exceeding 5,000 grams. This long-tail characteristic indicates that high emissions are usually associated with extreme situations such as long-distance travel or severe congestion. In subsequent modeling, logarithmic transformation or outlier processing of this variable can be considered to improve model stability.

The average speed also showed a right-skewed distribution, with the main peak at about 12 miles per hour, and most trips concentrated between 10 and 20 mph. This distribution reflects the low-speed driving environment common in New York City, which is consistent with traffic congestion or speed limit conditions, and high-speed trips are relatively rare.

In terms of environmental variables, PM2.5 concentrations are concentrated between 30 and 50 $\mu\text{g}/\text{m}^3$, with a peak of about 45 $\mu\text{g}/\text{m}^3$, indicating that most trips occur at moderate pollution levels, which is representative; while the average temperature is mainly distributed between 31 and 36 degrees Fahrenheit, reflecting that the data was mainly collected in winter, and the climate conditions are relatively stable, which is conducive to controlling the interference term of temperature in the model.

In addition, the time period variable shows that the trip volume is highest in the morning (6:00–12:00), followed by the afternoon, evening, and late night, indicating that the demand for online ride-hailing is most concentrated during the morning peak.

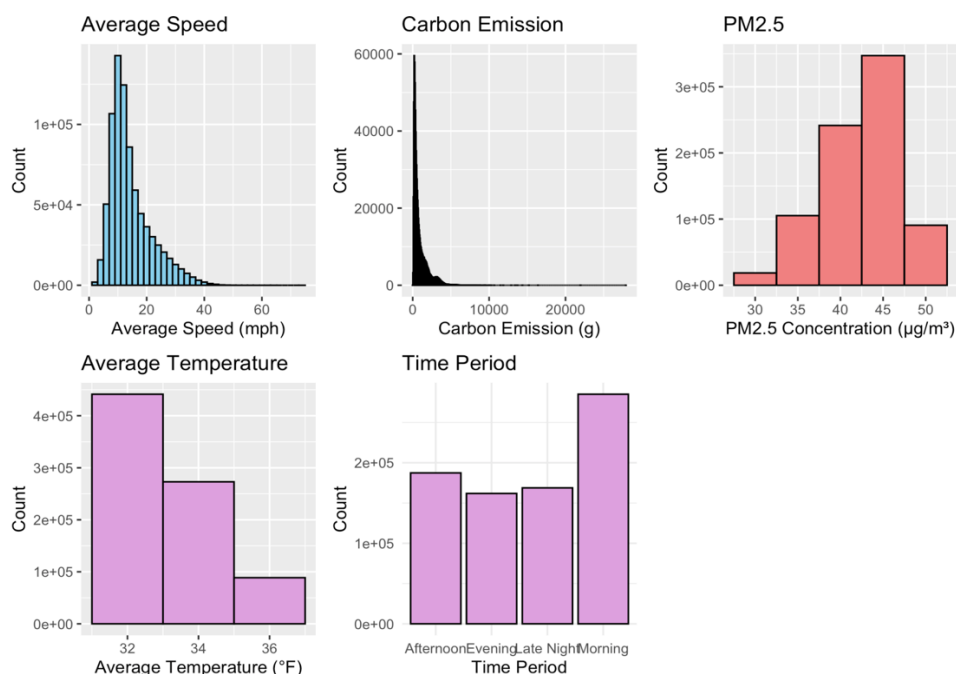


Fig1. Key variables of hvfhv

I then conducted a univariate exploratory analysis of traditional yellow taxi trips.

The distribution of Average Speed showed a clear right-skewed characteristic, with the vast majority of trips below 10 mph and very few exceeding 30 mph. This reflects that taxis mainly operate in traffic congestion or speed-limited areas, which is consistent with the actual conditions of urban roads in New York City.

The distribution of Carbon Emission also showed a strong long-tail characteristic, with emissions at low levels for the vast majority of trips and abnormally high values for a few trips. Such extreme emissions may be related to reduced efficiency due to long-distance travel or extremely low-speed operation, and logarithmic transformation or outlier processing should be considered in modeling.

In terms of environmental variables, the distribution of PM2.5 concentration is approximately normal, with the main peak concentrated between 30–40 $\mu\text{g}/\text{m}^3$, indicating that the overall air pollution during the data collection period was at a medium-to-high level and was representative. The average Temperature showed an irregular bimodal distribution, with two high-frequency intervals at approximately 35°F and 45°F, suggesting that the data may have been collected during the climate transition period such as late winter and early spring. Although the direct impact of temperature on emissions is limited, it may indirectly affect emission levels by affecting travel demand, driving behavior, and engine efficiency.

Finally, the statistical results of the Time Period show that taxi trips are mainly concentrated in the afternoon and evening, followed by the morning, and the least in the late night. This distribution reflects the obvious commuting peak and daily activity patterns.

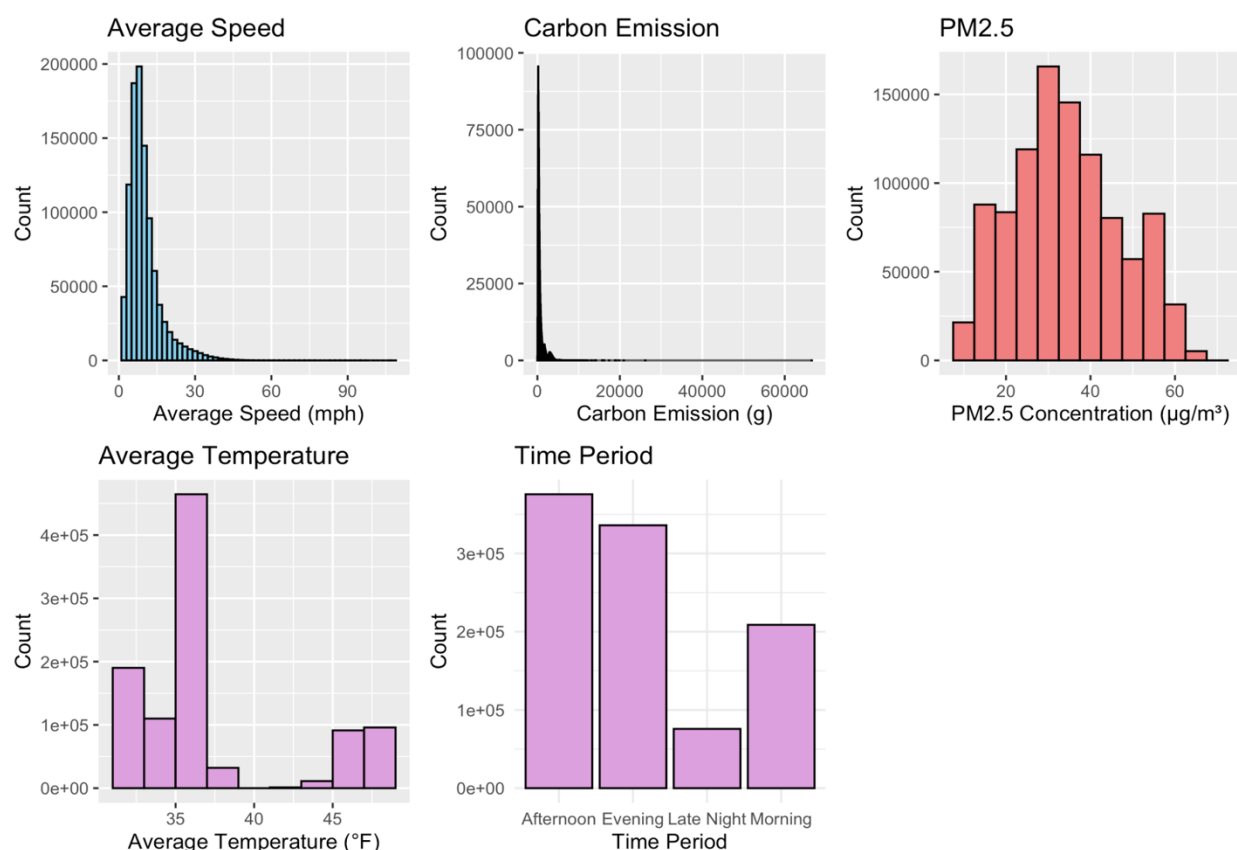


Fig2. Key variables of taxi

Bivariate or Multivariate Analysis

1. Carbon Emission vs trip distance

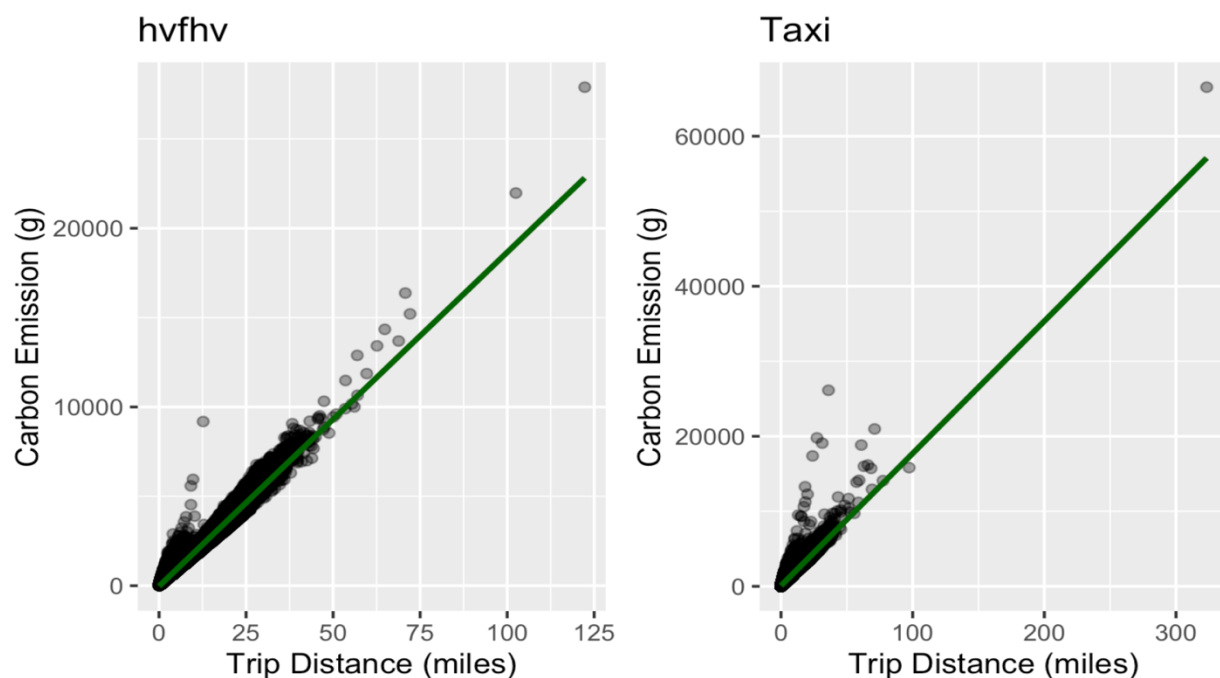


Fig3. Carbon emission vs. trip distance

It is obvious from Figure 3 that carbon emissions and travel distance show an approximately linear relationship: as the travel distance increases, carbon emissions also increase. This trend is consistent with the common sense emission mechanism, that is, when other conditions (such as vehicle speed, congestion level, and vehicle model) remain roughly unchanged, the longer the driving distance, the more fuel consumption, and the corresponding CO₂ emissions are also higher.

The data in the left figure (hvfhv) shows that the vast majority of trips are concentrated between 0-100 miles, the point cloud is densely distributed, and the linear relationship is stable, indicating that the growth trend of emissions with distance within the daily travel range is relatively consistent. In the right figure (taxi), although the overall linear relationship is still obvious, due to the inclusion

of extremely long-distance samples (such as more than 300 miles), some points deviate from the main trend.

2. PM2.5 in different Boroughs

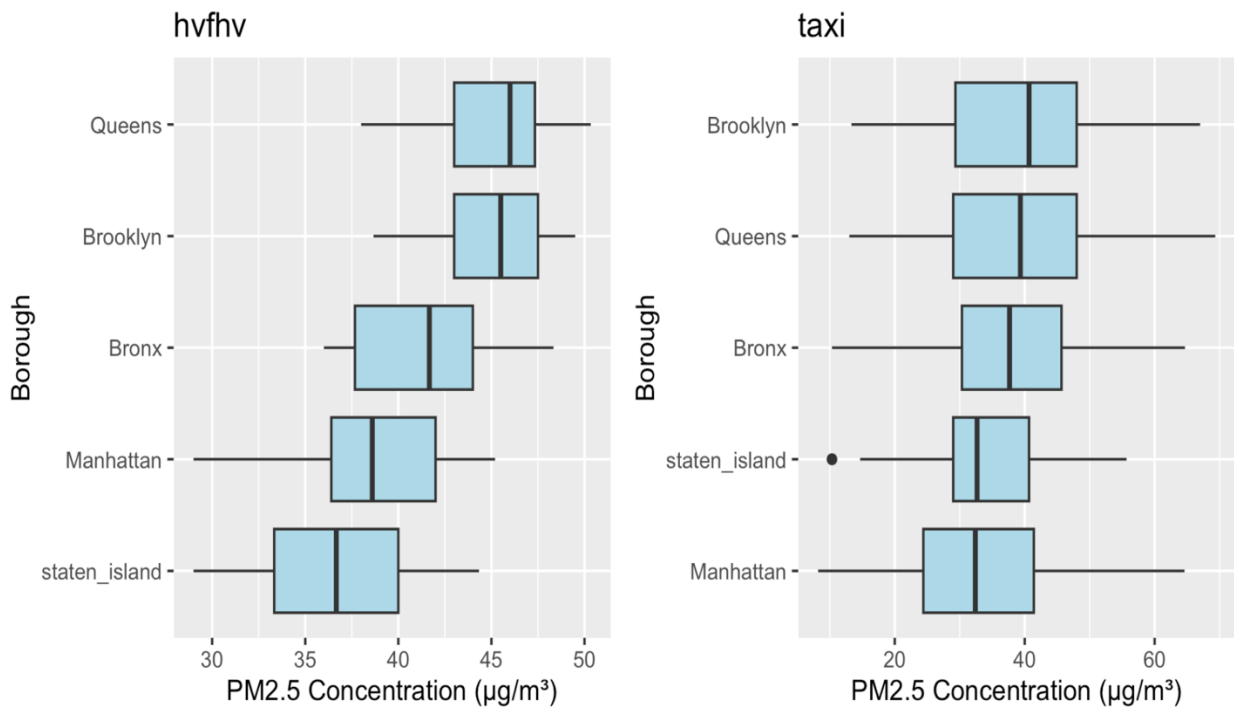


Fig4. PM2.5 in different areas

From Figure 4, both travel datasets show significant spatial heterogeneity, and there are systematic differences in air pollution levels in different administrative areas.

In the hvfhv data, the median PM2.5 in Queens and Brooklyn is significantly higher than that in other areas, with a distribution concentrated at 43–47 $\mu\text{g}/\text{m}^3$, indicating that these two areas may face a heavier burden of traffic pollution. Manhattan and the Bronx are next, while Staten Island has the lowest pollution level, with a relatively concentrated distribution around 37 $\mu\text{g}/\text{m}^3$. This trend may reflect the differences in vehicle density and traffic intensity between regions, especially Queens and Brooklyn as transportation hubs and densely populated areas, where higher pollution levels are expected.

In contrast, the distribution of PM2.5 in the taxi data is wider, especially in Queens and the Bronx, with a larger interquartile range, indicating greater fluctuations in air quality. The overall levels of Staten Island and Manhattan are relatively low. The low-tail outlier value of Staten Island deserves attention, which may be related to its low motor vehicle density and high green space coverage.

3. Average temperatures for different Boroughs

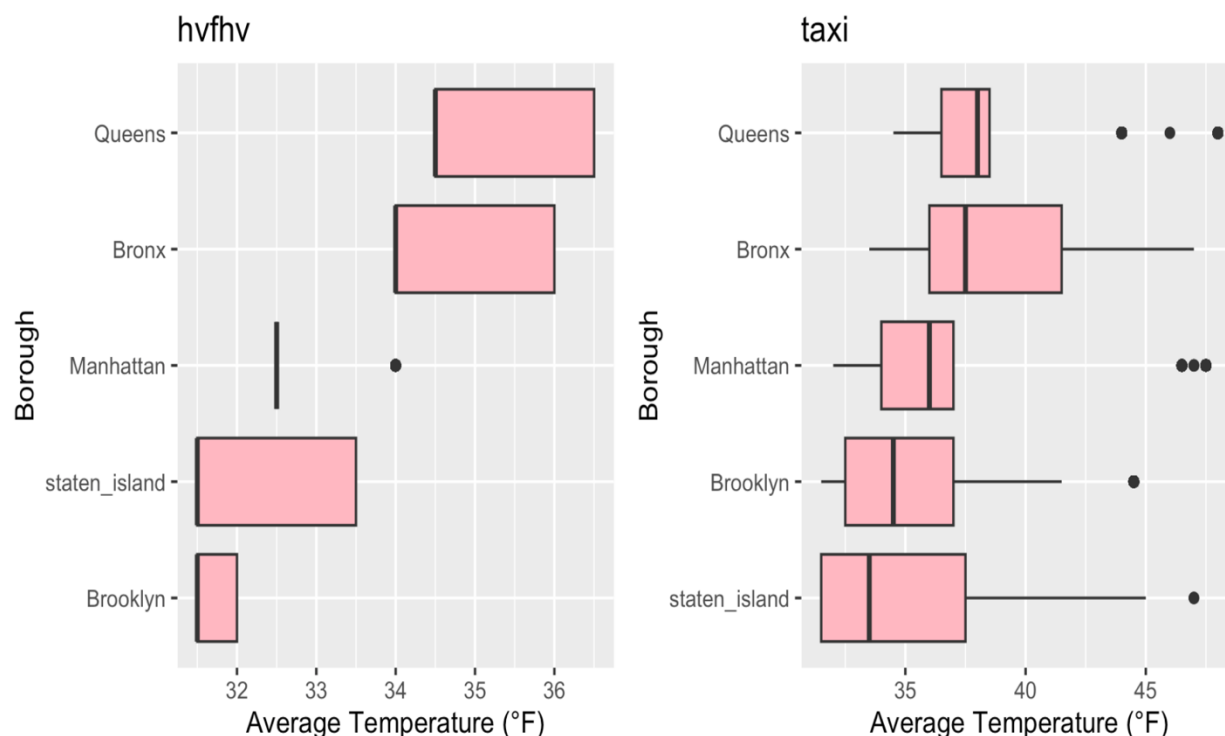


Fig5. The average temperature in different areas

Hvfhv data show significant differences in average temperatures across boroughs. Queens and the Bronx have median temperatures between 35–36°F, significantly higher than Brooklyn and Staten Island, which are concentrated in the 31–33°F range; Manhattan's temperature distribution is extremely concentrated, with almost no upper and lower quartile ranges, which may be due to the small sample size or short recording time.

Taxi data show a more stable temperature distribution. The intervals overlap significantly, indicating that the sampling time is relatively consistent, and the median temperature is mostly between 34–38°F. The Bronx has the widest range of temperature fluctuations, while other areas are relatively concentrated. Some areas (such as Queens and Manhattan) have significant high-temperature anomalies, which may be related to individual warm winter days or data errors.

Data Cleaning

Missing Values:

In terms of missing value processing, I did the following processing:

1. PM2.5: Matched with the EPA air quality dataset based on time and region; Considering that NO₂ and O₃ are seriously missing, they are not included in the modeling; at the same time, in order to control data consistency, only complete records within December 2024 are retained.
2. RatePerDistance and CarbonEmission_g: After a preliminary screening of missing records, it was found that a large number of corresponding travel distances were less than 1 mile, and it was speculated that there were anomalies, so they were uniformly eliminated.
3. Average Temperature: Matching and filling by referring to the meteorological dataset ensures that each record has temperature information.
4. Borough Information: To focus on the five core boroughs, records with missing areas or marked as other areas (such as EWR) were deleted to ensure consistency and comparability of spatial divisions.

Outliers:

In the original data, carbon emissions show a strong right-skewed distribution. Most records are concentrated in the low-value range, while a few extremely high values are far away from the main distribution. For linear regression or tree models, this skewness may not only affect the stability of parameter estimation but also amplify the impact of outliers on model fitting and reduce the

overall prediction accuracy. Therefore, I logarithmically transform the carbon emission data to alleviate the impact of extreme values and improve the model's ability to learn the overall data structure.

Specifically, I used the $\log1p()$ function to transform *CarbonEmission_g*. After the transformation, it can be observed that the logarithmic carbon emission values of most samples in the two data sets are concentrated around 5, and the originally sparse long-tail high values are also compressed to a more reasonable range, making the overall distribution more compact and smooth.

Transformation:

To further characterize the impact of periods on travel behavior and emissions, I added a time period classification variable *time_period* to each record based on the pickup time. The specific processing flow is as follows: extract the hours and divide them into four time periods: Late Night (0:00–6:00), Morning (6:00–12:00), Afternoon (12:00–18:00), and Evening (18:00–24:00). This variable will be included in the model as a categorical feature to capture the potential differences in travel patterns and carbon emission behaviors in different time periods.

Train-Test Split

In order to build a unified and comparable carbon emission prediction model, I merged the two datasets into one overall dataset. Although the two represent different types of travel modes, they are highly consistent in structure and have the same core predictive variables, such as travel distance, speed, environmental variables, etc. Merging the two for modeling helps to increase the sample size, enrich data variability, and improve the generalization ability of the model in different traffic scenarios.

In terms of data partitioning, I use an 80%-20% method to divide the training set and the test set to ensure that enough data is retained for parameter learning when training the model while retaining independent samples for evaluating the performance of the model on unseen data.

Modeling Strategy and Results

Ridge Regression and Lasso Regression

To solve the potential multicollinearity problem and improve the stability and interpretability of the model, I first introduce two regularized linear regression models: the Ridge regression and the Lasso regression. They effectively suppress overfitting by introducing L2 and L1 regular terms in the loss function to realize parameter shrinkage and variable selection, respectively.

In the modeling process, cross-validation was used to select the optimal regularization coefficients, and the log-transformed carbon emission values were used as the prediction targets. The test set results show that:

1. The performance of Ridge Regression is $RMSE = 0.385$, $MAE = 0.290$, $R^2 = 0.832$
2. Lasso Regression performs better with $RMSE = 0.373$, $MAE = 0.269$, $R^2 = 0.842$

This suggests that Lasso is more suitable for the current data structure, and can simultaneously improve prediction ability and provide sparse interpretation, so we choose it as a baseline model to provide a reference for subsequent complex models.

Random Forest

In order to further capture the possible nonlinear relationship between features, an ensemble learning method, Random Forest, was introduced. This model can effectively reduce the high variance problem of a single tree by constructing a large number of decision trees and averaging them.

As an integrated learning method, randomForest has good fitting ability and robustness by constructing multiple decision trees and making integrated predictions. However, in the early stage

of model tuning, due to the huge amount of data, the problem of memory overrun or computational interruption occurs when trying to use the standard randomForest package several times. For this reason, I switched to the more computationally efficient ranger package to build the model.

Based on the ranger implementation, I systematically tuned the model hyperparameters through cross-validation. The parameter tuning goal is to minimize the RMSE of cross-validation. The final results show that the optimal parameter combination is: *mtry* = 5, *splitrule* = "variance", *min.node.size* = 5. Among them, *mtry* controls the number of variables randomly selected for each split, *splitrule* determines the splitting criterion, and *min.node.size* limits the minimum number of samples in the leaf node of each tree. These parameters work together to suppress overfitting while maintaining the model's expressiveness.

The performance on the test set is good: RMSE = 0.0052, MAE = 0.0019, $R^2 = 0.99997$. Although this result is extremely ideal, it also indicates the potential risk of overfitting.

XGBoost

Finally, the XGBoost model was introduced to improve prediction flexibility and model interpretability.

Considering that the total amount of our merged data set is close to 1.4 million, if we directly perform parameter tuning on the full amount of data, we will face practical challenges such as high consumption of computing resources and long running time. Therefore, in the parameter tuning stage, I adopted the strategy of randomly extracting 1% of samples from the original data for testing experiments. On this basis, we conducted a grid search, and the main adjustment parameters included learning rate (*eta*), maximum tree depth (*max_depth*), minimum number of leaf node samples (*min_child_weight*), etc. In the end, the model performed best under the parameter combination of *eta* = 0.05, *max_depth* = 8, and *min_child_weight* = 1. The cross-validation result RMSE was 0.0152 and the optimal round was 500. This combination achieved a good balance in learning speed and model complexity.

After obtaining the optimal parameters, I applied them to the complete training set for modeling. The final performance of the model on the test set is: RMSE = 0.0051, MAE = 0.0032, $R^2 = 0.99997$, which is almost the same as the random forest. This result shows that XGBoost can fit the logarithm of carbon emissions with high precision, while having higher regulation flexibility and model interpretability. Since we introduced the early stopping strategy during the training process, once the error of the validation set no longer decreases, the training is stopped, which effectively avoids overfitting and makes the model also show good robustness on the actual test data.

In summary, XGBoost is regarded as the modeling scheme with the most comprehensive advantages in this study due to its powerful nonlinear expression ability, adjustable parameter mechanism and good generalization performance.

Conclusion

In this study, I integrated the travel data of traditional taxis and online ride-hailing cars, divided them into training sets and test sets, and used three models, Lasso regression, Random Forest, and XGBoost, to predict the logarithmically transformed carbon emissions. Model selection covers different strategies from linear to nonlinear, from sparse interpretation to high-precision fitting, to comprehensively evaluate the performance of various methods in this task.

From the results, the Lasso model achieves feature screening and overfitting control through L1 regularization while retaining linear interpretability, with R^2 reaching 0.842, and has good basic performance. In contrast, Random Forest greatly improves the model's fitting ability by integrating multiple decision trees, with R^2 as high as 0.99997 on the test set, but there is also a certain risk of overfitting. While ensuring high precision, XGBoost enhances generalization ability through the early stopping strategy, with R^2 reaching 0.99997, which is almost the same as Random Forest,

and has stronger adjustability and feature importance output capabilities. Therefore, we believe that XGBoost achieves a good balance between prediction performance and model controllability.

Further analyzing the feature importance of the model, we found that trip distance (*trip_miles*) and trip time (*trip_time*) are the most critical variables for predicting carbon emissions in both random forests and XGBoost. This finding is consistent with the mechanism of motor vehicle emissions: carbon emissions depend largely on the distance and time traveled. Average speed (*avg_speed_mph*) is the second most important. Although it contributes a small gain in XGBoost, it has a high split frequency, indicating that it frequently participates in the decision-making process in the model. In addition, the influence of administrative district information (*pickup_borough*) and travel mode (*source*) in both models is small, indicating that their marginal contribution to emissions is limited after controlling other variables.

In summary, this study established a framework for urban carbon emissions prediction based on multi-source travel data and clarified the most suitable modeling strategy through model comparison. The XGBoost model has become a recommended application model due to its high prediction accuracy and robustness. At the same time, the feature importance analysis verifies the dominant role of travel behavior variables in emissions, providing a clear direction for future system deployment and mechanism optimization.

These predictions are not only valuable at the technical level, but also have broad significance in practical applications. First, integrating carbon emission prediction results in the navigation system can achieve low-emission route recommendations and promote green travel for the public. Secondly, governments and transportation departments can use such models to identify and regulate high-emission periods and areas and provide data support for air quality improvement policies. In addition, platform companies can also use carbon emission modeling results to evaluate the environmental impact of operations and better connect ESG indicators with carbon disclosure requirements. More importantly, this type of data-driven system can guide the public to understand the carbon footprint of their own travel, promote personal environmental responsibility, and thus promote the transformation of urban transportation systems towards a more low-carbon and intelligent direction.